

LAKE CUMBERLAND, KY, USA

WMO: 724354

Lat: 37.054N

Lon: 84.615W

Elev: 927

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 63815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	17.7	1.0	6.0	16.6	6.8	8.1	20.1	21.2	48.8	18.9	47.3	3.3	0	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	93.2	74.2	90.8	73.6	89.1	72.9	77.5	89.1	76.1	86.6	74.8	84.4	5.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	128.7	81.3	72.9	126.3	80.4	72.2	123.5	79.5	41.6	88.1	40.1	86.7	38.8	84.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.8	15.4	12.6	DB	6.5	95.7	7.5	4.8	1.0	99.1	-3.4	102.0	-7.6	104.7	-13.1	108.2	(4)
			WB	4.7	78.7	7.4	2.0	-0.6	80.2	-4.9	81.4	-9.1	82.5	-14.4	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	37.1	40.7	47.7	57.7	66.2	73.7	76.7	75.8	69.9	58.5	47.2	40.0	
	DBStd	16.49	11.71	10.93	9.96	9.03	7.22	5.20	4.18	4.88	6.45	8.46	9.20	10.16	
	HDD50	1442	427	294	165	32	1	0	0	0	0	22	160	341	
	HDD65	3992	866	680	539	249	77	5	0	1	28	234	537	776	
	CDD50	4252	27	34	94	263	502	712	829	801	598	286	75	31	
	CDD65	1328	1	1	4	30	113	267	364	337	176	32	3	0	
	CDH74	12048	1	2	46	327	1014	2408	3233	3038	1625	333	20	1	
	CDH80	4642	0	0	4	59	279	932	1341	1322	631	73	1	0	
Wind	WSAvg	4.2	5.5	5.4	5.6	5.3	4.1	3.5	2.9	2.7	2.8	3.4	4.2	4.7	
Precipitation	PrecAvg	49.6	4.7	3.9	5.0	4.0	4.9	4.6	4.5	3.8	3.0	3.1	3.3	4.7	
	PrecMax	61.5	8.7	6.9	11.3	9.0	10.8	10.2	8.1	7.8	6.0	6.3	7.3	13.6	
	PrecMin	36.6	1.7	1.6	2.1	1.2	2.2	1.8	1.6	1.0	0.2	0.4	0.8	1.4	
	PrecStd	7.3	1.6	1.5	2.4	1.7	2.2	2.2	1.6	1.7	1.7	1.7	1.5	2.6	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	72.6	79.4	85.5	89.8	93.4	96.8	98.4	96.5	87.6	77.4	70.0
MCWB			60.1	58.5	60.7	66.7	70.9	75.3	76.4	74.2	72.0	69.3	61.6	60.2	
2%		DB	63.7	67.7	73.3	81.3	86.1	90.8	92.1	93.2	90.6	81.5	72.1	63.9	
		MCWB	56.9	57.4	58.4	62.7	69.0	73.9	75.4	74.0	70.5	65.0	58.9	57.2	
5%		DB	60.6	63.2	69.9	78.5	82.4	88.4	90.2	90.5	87.7	78.5	67.1	60.7	
		MCWB	54.8	53.2	56.8	61.2	67.5	72.7	74.7	73.8	70.2	63.2	57.4	54.9	
10%		DB	54.9	58.6	64.2	73.5	80.7	85.9	87.8	87.9	83.5	73.0	63.2	55.3	
		MCWB	48.8	50.1	53.4	59.4	66.5	71.2	73.8	73.0	68.5	61.7	54.8	50.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	61.3	61.9	63.8	68.5	73.8	78.3	80.1	78.8	75.6	71.9	65.9	62.9
			MCDB	66.5	68.2	74.6	81.6	84.9	91.0	93.4	91.2	87.7	80.2	73.1	66.4
	2%	WB	58.6	58.9	60.9	65.8	71.4	76.2	77.9	77.2	73.5	69.1	62.4	59.1	
		MCDB	63.4	65.7	70.2	76.6	81.4	87.3	89.1	88.9	84.0	75.8	68.6	63.1	
	5%	WB	55.2	55.2	58.4	63.5	69.6	74.6	76.3	75.5	72.1	66.5	59.3	55.2	
		MCDB	59.5	60.4	66.8	73.2	78.9	84.6	86.3	86.0	81.8	73.7	64.3	59.1	
	10%	WB	50.4	51.7	55.4	61.5	68.0	73.0	75.0	74.1	70.7	64.0	55.8	51.9	
		MCDB	54.3	57.0	62.9	70.9	76.9	82.3	83.9	83.5	79.4	70.3	61.5	55.6	
	Mean Daily Temperature Range	5% DB	MDBR	17.2	18.9	21.3	23.3	21.7	20.9	19.8	21.4	22.9	23.4	20.7	16.7
			MCDBR	20.7	25.9	28.1	28.0	25.5	24.6	23.4	25.8	29.0	28.9	26.6	20.7
MCWBR			15.1	16.8	16.1	13.8	11.3	10.2	10.0	10.7	11.6	14.2	15.7	14.6	
5% WB		MCDBR	18.5	20.6	22.6	22.1	21.2	20.7	19.9	21.5	21.9	21.5	19.7	18.1	
		MCWBR	14.6	16.0	14.5	12.5	10.1	9.6	9.2	9.5	9.7	11.9	14.4	14.3	
Clear-Sky Solar Irradiance	taub	0.309	0.319	0.352	0.396	0.448	0.465	0.496	0.499	0.414	0.361	0.323	0.305		
	taud	2.495	2.457	2.407	2.281	2.167	2.156	2.089	2.066	2.287	2.428	2.499	2.518		
	Ebn at Noon	280	291	290	281	267	261	252	248	265	271	272	274		
	Edh at Noon	26	30	35	42	47	48	51	51	39	30	25	23		
All-Sky Solar Radiation	RadAvg	646	844	1203	1590	1770	1951	1829	1728	1442	1111	788	565		
	RadStd	70	119	108	117	158	108	96	91	152	142	85	56		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (20 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+153	

Nomenclature: See separate page